

1) Does the treatment of a stress fracture in a foot affect the success or failure of healing the bone? A recent experiment in a medical journal took four separate random samples of various treatment methods used to treat a fractured foot. In each of these random samples, they recorded whether the patient saw success in the healing of the fracture. A Chi-Square test for Homogeneity was performed and the follow-up analysis is given below.

	Success	Failure
Surgery	54 50.471 0.247	12 15.529 0.802
Weight-Bearing Cast	41 51.235 2.045	26 15.765 6.645
Non-Weight Bearing Cast for Less Than 6 Weeks	17 19.118 0.235	8 5.882 0.762
Non-Weight Bearing Case for 6 Weeks	70 61.176 1.273	10 18.824 4.136

Key: Observed Expected Contribution
--

- What would be the null and alternative hypotheses in this situation?
- Show how the value of “15.529” under “surgery, failure” was obtained.
- What would be the chi-square statistic and p-value for this test?
- What is your conclusion in the context of the problem, at the 1% significance level?

e) Identify the two largest contributions to the chi-square statistic. What do these contributions imply in the context of the problem?

2) A medical researcher is investigating the effectiveness of five different treatments (A, B, C, D, and E) on the recovery outcomes of patients with a specific illness. The recovery outcomes are categorized as Excellent, Good, No Improvement, Not Good, and Poor. The data collected from a clinical trial involving 425 patients is presented in the table below.

Observed Counts	Excellent	Good	No Improvement	Not Good	Poor
A	18	22	17	23	5
B	20	25	15	10	15
C	22	20	10	13	15
D	15	30	25	15	5
E	25	15	21	14	10

a) What are the null and alternative hypotheses for this chi-square test for homogeneity?

b) Using your calculator, find the expected counts, chi-square statistic, degrees of freedom, and the p-value.

Expected Counts	Excellent	Good	No Improvement	Not Good	Poor
A					
B					
C					
D					
E					

$\chi^2 =$ _____

df = _____

P-value = _____

c) Based on a significance level of 1%, what can you conclude?